

Multimodal Deep Learning for Proactive Wildfire Burned Area Prediction at Ignition

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Abstract

Wildfires cause a significant global threat, making prediction at the moment of ignition critical for timely intervention. This work presents a multimodal machine learning framework for forecasting the final burned-area extent using only pre-fire environmental data. The task is formulated as a semantic segmentation problem based on satellite imagery (Sentinel-2 and Landsat), meteorological variables from ERA5, and topographic features derived from a Digital Elevation Model (DEM). The proposed approach employs a Multi-Modal Feature Pyramid Network (MM-FPN) with feature-level fusion and incorporates an auxiliary land-cover segmentation task to evaluate the effect of multi-task learning. Using a conservative decision threshold of 0.95 to reduce false positives, the multimodal model improves Intersection over Union (IoU) from 0.15 in the single-task baseline to approximately 0.40. These results demonstrate that multimodal fusion combined with auxiliary semantic supervision enables more accurate early burned-area estimation and supports informed decision-making during initial wildfire response.

1 Introduction

Wildfires represent a growing environmental and social challenge in regions such as Piedmont, Italy, where the frequency and severity of fires have increased in recent years[8]. Although fire risk assessment has been widely studied, most existing approaches remain reactive, relying on post-ignition models that require active fire perimeters to estimate spread [5]. As a result, they do not support decision-making during the "Golden Hour". To address this limitation, we propose a multimodal deep learning framework for proactive burned area delineation that operates at the moment of ignition using only pre-fire environmental data. The problem is formulated as a semantic segmentation task and implemented using a Multi-Modal Feature Pyramid Network (FPN) to integrate heterogeneous inputs, including satellite imagery (Sentinel-2, Landsat), topographical information (DEM), and meteorological variables from ERA5 [7]. This design enables the model to capture complex, non-linear interactions between environmental conditions and potential fire spread. In addition, we examine the role of Multi-Task Learning (MTL) in improving prediction robustness[4]. In addition to the primary burned area segmentation task, an auxiliary land cover segmentation objective is introduced. By encouraging the shared

encoder to learn terrain semantics relevant to fire behavior, this auxiliary supervision promotes better generalization of early burned-area predictions[3, 4].

1.1 Research Questions

We evaluate our approach in a comprehensive dataset of historical wildfires in the Piedmont region (2012–2024). Based on these settings, we address the following research questions:

- **RQ1:** Can multimodal pre-fire environmental data enable accurate burned-area prediction at the moment of ignition?
- **RQ2:** Does auxiliary-task learning improve spatial segmentation performance compared to single-task models?
- **RQ3:** What is the relative contribution of satellite, meteorological, and topographical modalities?

The remainder of this paper is organized as follows: we review related work, describe the proposed approach, and then present the experimental setup and results, followed by a concluding discussion. All sources are available on the following public GitHub repository: 2025-P8-Fire-Ignition-Maps.

2 Related Work

Wildfire risk assessment has traditionally relied on physical models and data-driven approaches that estimate fire danger from meteorological variables or post-ignition observations[1]. Although recent deep learning methods achieve strong performance in active fire detection[5] and mapping of burned areas[11], they remain largely reactive, requiring thermal anomalies or active fire perimeters. Consequently, these approaches offer limited support for decision-making during the early stages of wildfire development[10]. In contrast, our work focuses on proactive prediction by forecasting the final burned area at the moment of ignition using only pre-fire environmental data.

Recent advances in semantic segmentation have enabled pixel-level environmental modeling using deep convolutional networks[2]. However, studies consistently show that optical imagery alone is insufficient for predictive wildfire modeling, as fire behavior depends on complex interactions between meteorology, topography, and vegetation structure[6]. Motivated by this limitation, we formulate burned area prediction as a multimodal feature fusion problem and employ a Feature Pyramid Network to integrate

heterogeneous environmental inputs across multiple spatial scales[7].

Multi-Task Learning (MTL) has been shown to improve generalization by jointly optimizing related objectives within a shared encoder[4]. In remote sensing, auxiliary tasks such as land cover segmentation encourage the learning of semantically meaningful terrain representations and act as a regularizer, particularly in sparse or imbalanced datasets[3]. Building on this insight, we evaluate the contribution of auxiliary land cover supervision by comparing single-task and multi-task multimodal models under identical encoder configurations.

3 Methodology

3.1 Problem Formulation

We formulate burned area mapping as a supervised multimodal semantic segmentation problem. Let $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^N$ denote a dataset of spatially aligned samples over a region $\Omega \subset \mathbb{R}^2$. Each input X_i consists of heterogeneous pre-fire modalities, including optical satellite imagery (Sentinel-2 and Landsat-8), static geospatial features, and meteorological variables derived from ERA5 data.

The primary objective is to learn a mapping function f_θ that predicts a binary burned area mask $Y^{(BA)} \in \{0, 1\}^{H \times W}$. To enforce semantic consistency and constrain the learned feature space, an auxiliary task is introduced to predict a multi-class land cover map $Y^{(LC)} \in \{0, \dots, C-1\}^{H \times W}$ with $C = 12$ classes.

The model jointly optimizes the conditional distribution

$$p(Y^{BA}, Y^{LC} | X; \theta) = p(Y^{BA} | H; \theta_{BA}) p(Y^{LC} | H; \theta_{LC}). \quad (1)$$

where H denotes a shared latent representation learned from fused multimodal inputs, and θ_{BA} and θ_{LC} correspond to task-specific decoding parameters.

3.2 Architecture

We propose a Multi-Modal Feature Pyramid Network (MM-FPN) to integrate heterogeneous data sources with differing spatial resolutions and physical characteristics. The architecture shown in Figure 1 follows a feature-level fusion strategy in which high-resolution Sentinel-2 imagery serves as the primary backbone, while auxiliary modalities are incorporated at multiple semantic scales. This design prioritizes spatial detail from optical imagery while allowing complementary environmental context to modulate predictions without diluting fine-grained structure.

Sentinel-2 inputs are encoded using an EfficientNet-B4 backbone, producing a hierarchy of multi-scale feature maps that capture both local spatial detail and higher-level semantic information. Auxiliary data sources, including Landsat-8 imagery, static geospatial rasters, and ERA5 meteorological variables, are spatially aligned and normalized, then projected into the Sentinel feature space using lightweight convolutional or linear transformations. These projected features are adaptively resized and fused with the corresponding Sentinel encoder outputs at each scale through a weighted cross-modal fusion mechanism with

channel and spatial attention, enabling context-aware integration while maintaining computational efficiency.

The fused feature pyramid is decoded using a Feature Pyramid Network (FPN) to recover high-resolution spatial predictions. The decoder branches into a primary binary head for burned-area segmentation and an auxiliary multi-class land-cover head used only during training to regularize feature learning. During inference, predictions rely exclusively on the burned-area output.

3.3 Multimodal & Multi-Task Learning

The proposed framework integrates heterogeneous pre-fire data through a feature-level multimodal fusion strategy that combines complementary environmental cues while preserving high-resolution optical information. After spatial alignment, auxiliary modalities are projected into the Sentinel-2 feature space using lightweight transformations and fused with Sentinel representations at multiple encoder stages via a weighted cross-modal mechanism with channel and spatial attention.

To regularize learning, the model adopts a multi-task formulation that jointly optimizes burned-area segmentation and land-cover segmentation. Both tasks share the same encoder-decoder architecture and differ only in their output heads. The auxiliary land-cover task is used exclusively during training to encourage semantically meaningful terrain representations and improve physical consistency in burned-area predictions.

4 Experimental Setup

This section outlines the experimental framework used to evaluate the proposed model.

4.1 Dataset

The study focuses on the Piedmont region [9], which exhibits heterogeneous landscapes and complex topography. The proposed Piedmont-2012–2024 dataset covers a twelve-year period and integrates five spatially and temporally aligned data modalities to support multimodal learning.

Sentinel-2 multispectral imagery serves as the primary input modality, with twelve spectral bands retained per event and channel-wise normalization applied using training-set statistics. Complementary optical and thermal information from Landsat-8 is incorporated and resampled to the Sentinel-2 spatial resolution. Meteorological context is provided by ERA5 reanalysis variables, including temperature and wind components, which are projected and aligned with the optical inputs to supply global environmental context. Static geospatial features, including a Digital Elevation Model (DEM) and street-density maps, are used to represent topographic and anthropogenic influences.

The dataset provides two types of pixel-level annotations. Primary labels consist of binary burned-area masks derived from regional fire databases and FIRMS, while auxiliary labels correspond to multi-class land-cover maps used for secondary semantic supervision.

To ensure realistic evaluation and avoid temporal leakage, data are split chronologically: events from 2017–2021

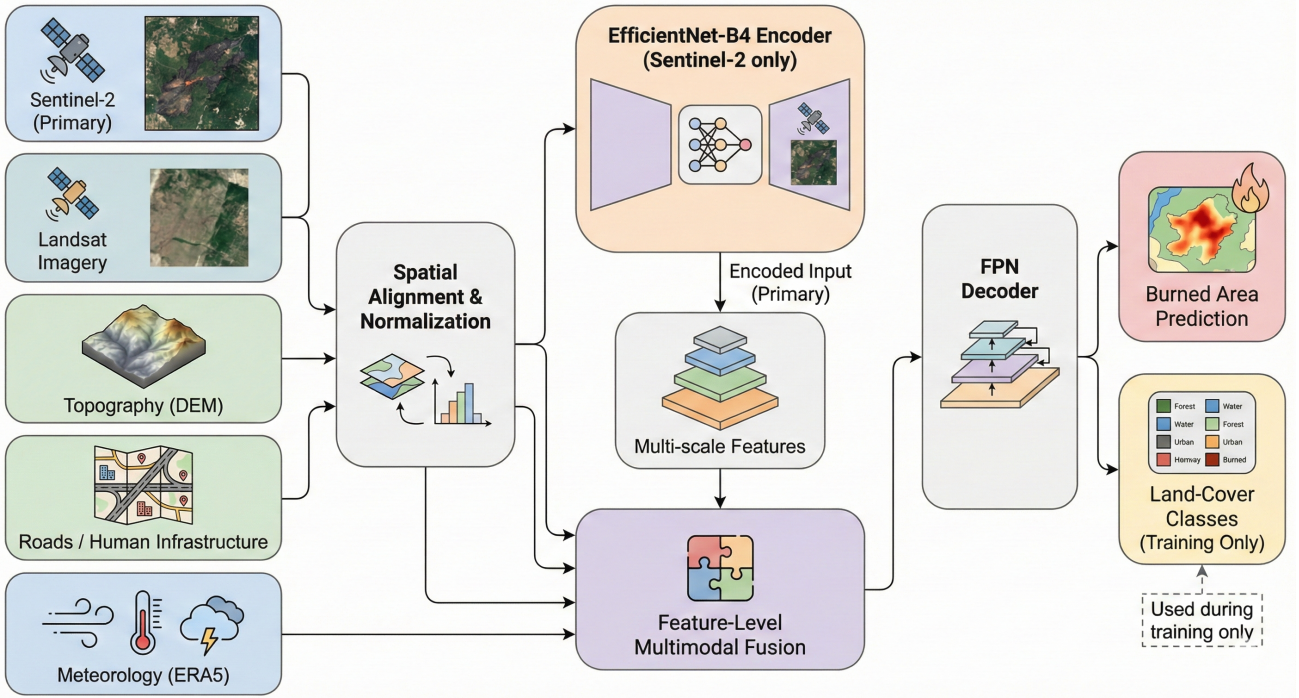


Figure 1: Multi-Modal Feature Pyramid Network (MM-FPN): pre-fire satellite, topographical, and meteorological data are fused prior to encoding to produce burned-area segmentation, with an auxiliary land-cover segmentation head.

are used for training, 2022 for validation and model selection, and 2023 is reserved as a held-out test set to evaluate generalization to unseen wildfire seasons.

4.2 Training Details

The model was implemented in PyTorch and trained on NVIDIA GPU RTX 4070 with paired 12 GB GDDR6X memory hardware. Sentinel-2 imagery was processed using an EfficientNet-B4 encoder initialized with ImageNet weights, providing multi-scale features for a feature pyramid network decoder. Auxiliary modalities, including Landsat imagery, ERA5 variables, and static geospatial layers, were projected into the Sentinel feature space and fused with encoder features at multiple scales before decoding.

4.3 Multimodal Fusion Strategy

To combine heterogeneous data sources, the model employs a weighted feature-level fusion strategy. Based on prior ablation analyses assessing the relative information content of each modality, channel-wise fusion weights are initialized to fixed empirical values: $[1.0, 1.0, 0.42, 0.50, 0.33]$ for Sentinel-2, Landsat-8, DEM, street density, and ERA5, respectively. This design prioritizes high-resolution optical and thermal features, while contextual environmental information is adaptively integrated through channel and spatial attention mechanisms within the fusion blocks.

4.4 Optimization and Loss Function

Training is conducted using the AdamW optimizer with a multi-task loss objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{burn}} + \lambda \cdot \mathcal{L}_{\text{aux}} \quad (2)$$

The burned-area loss combines Binary Cross-Entropy and Dice loss with equal weighting to mitigate severe class imbalance. The auxiliary land-cover task is optimized using Cross-Entropy loss, with $\lambda = 0.1$ controlling its contribution to regularize the shared representation without dominating the primary objective.

4.5 Hyperparameters

Training is performed with a batch size of 8 and an initial learning rate of 1×10^{-4} , which is reduced using a ReduceLROnPlateau scheduler (factor 0.5, patience 5). Early stopping is applied with a patience of 15 epochs. Data augmentation includes synchronized geometric transformations, such as random flips and rotations, applied consistently across all spatial inputs and masks, while photometric augmentations, including random brightness and contrast adjustments, are applied only to the Sentinel-2 optical bands.

4.6 Evaluation Metrics

Evaluation focuses on burned-area segmentation performance. To address severe class imbalance, we report metrics robust to background dominance, using Intersection over Union (IoU) as the primary metric alongside precision and recall. To assess the impact of auxiliary supervision, we compute ΔIoU as the per-sample IoU difference between the multi-task model and a single-task baseline, aggregated by dominant land-cover class. Validation metrics are computed at a fixed decision threshold, while final evaluation includes threshold-independent precision-recall analysis to identify the optimal inference operating point.

Table 1: Quantitative comparison between the single-task baseline and the proposed multimodal model on the 2023 test set. Best-threshold metrics correspond to $\tau = 0.95$.

Metric	Baseline	Multimodal
IoU ($\tau = 0.95$)	0.1515	0.4026
F1-score ($\tau = 0.95$)	0.2632	0.5741
Precision ($\tau = 0.95$)	0.3468	0.5761
Recall ($\tau = 0.95$)	0.2120	0.5722
IoU ($\tau = 0.50$)	0.1515	0.2341
F1-score ($\tau = 0.50$)	0.2632	0.3793

4.7 No-Aux (single-task) baseline:

To isolate the contribution of auxiliary supervision, we train a controlled single-task baseline that preserves the *same* MultiModal FPN architecture, modality fusion strategy, and input protocol as the proposed model. It uses a ResNet-34 encoder initialized with ImageNet weights and is optimized solely for burned-area segmentation by disabling the auxiliary loss (i.e., $w_{\text{aux}}=0$, $w_{\text{mask}}=1$).

4.8 Overall Quantitative Performance

To address **RQ1**, we compare a single-task Sentinel-2 baseline against the proposed multimodal multi-task framework. Performance is evaluated on the held-out 2023 test set using metrics robust to severe class imbalance, with Intersection over Union (IoU) as the primary indicator of spatial accuracy.

At the operational decision threshold of $\tau = 0.95$, the multimodal model achieves an IoU of 0.4026, compared to 0.1515 for the baseline, corresponding to an improvement of approximately 70%. Consistent gains are observed across F1-score, precision, and recall, indicating that the performance improvement reflects a substantial enhancement in the learned representation rather than threshold effects alone. In particular, recall increases from 0.21 to 0.57, demonstrating improved recovery of burned pixels at ignition while maintaining high precision.

Figure 2 provides a qualitative comparison illustrating these gains. The multimodal prediction exhibits substantially improved spatial alignment with the ground-truth fire perimeter compared to the baseline, particularly in topographically complex regions. Together, the quantitative and qualitative results confirm that incorporating multimodal pre-fire environmental context significantly improves early burned-area estimation, supporting accurate prediction at the moment of ignition.

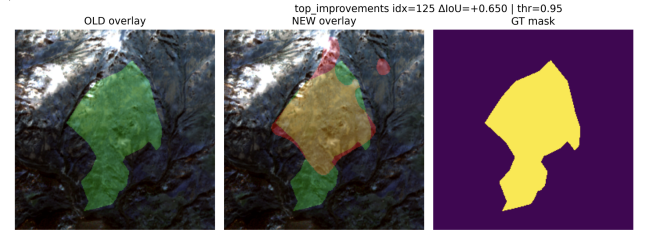


Figure 2: Qualitative comparison between the single-task baseline and the proposed multimodal model at $\tau = 0.95$. The multimodal prediction shows improved spatial alignment with the ground-truth burned area, particularly in complex terrain, resulting in a substantial per-sample IoU improvement ($\Delta\text{IoU} = +0.65$).

4.9 Effect of Decision Threshold on Spatial Predictions

The calibration of the decision threshold τ is pivotal in managing the trade-off between spatial conservative bias and over-prediction. As shown in the pixel-level sweep (Figure 3), peak IoU is achieved at a high threshold of $\tau = 0.95$, suggesting that the model produces highly polarized probability maps. At this threshold, the model effectively filters low-confidence noise in unburned regions, though it introduces a conservative bias that prioritizes high-confidence fire cores over peripheral edges. This results in improved precision by minimizing false positives on dark surfaces, but inherently increases false negatives—missing the extreme boundaries of the fire scar where environmental drivers may be superseded by active suppression efforts. Qualitative overlays in Figure 2 confirm that $\tau = 0.95$ provides the most reliable alignment for operational resource prioritization.

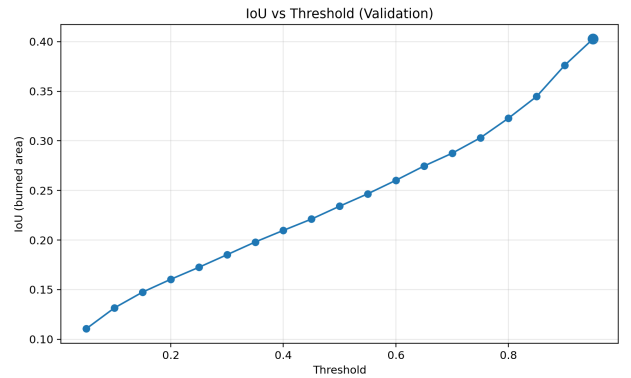


Figure 3: Threshold Sweep and Metrics

4.10 Precision–Recall Trade-off and Operating Point Selection

Figure 4 illustrates the precision–recall behavior of the proposed model on the validation set under a sweep of decision thresholds. As expected in a highly imbalanced burned-area segmentation setting, relaxing the decision threshold increases recall while progressively degrading precision due to over-prediction in unburned regions. Conversely, higher thresholds yield more conservative predictions, substantially reducing false positives at the cost of

missing low-confidence burned pixels, particularly along fire peripheries.

This trade-off is especially relevant for early-stage wildfire response, where false alarms can lead to inefficient allocation of limited suppression resources. In this context, the selected operating threshold of $\tau = 0.95$ corresponds to a high-precision regime of the precision-recall curve, emphasizing spatial reliability and confidence over exhaustive boundary coverage. While this choice sacrifices recall, it aligns with the objective of producing actionable burned-area estimates that prioritize correctness during the critical initial phase of wildfire management.

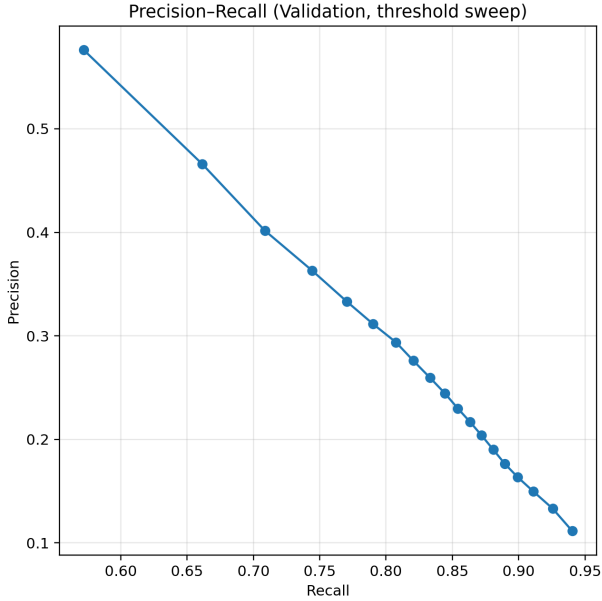


Figure 4: Precision-recall trade-off on the validation set under a sweep of decision thresholds. Higher thresholds favor conservative, high-precision predictions, while lower thresholds increase recall at the expense of spatial reliability.

4.11 Impact of Multi-Task Learning (MTL)

To address **RQ2**, we examine the effect of auxiliary land-cover supervision on burned-area segmentation performance. Adding the land-cover prediction task provides additional guidance during training, encouraging the shared representation to capture meaningful geographic context. This auxiliary supervision helps constrain burned-area predictions to locations that are physically plausible, such as excluding water bodies and urban regions.

Qualitative analysis of validation samples shows that the multi-task model consistently reduces false positives in heterogeneous landscapes, particularly in areas with mixed vegetation, bare soil, and human infrastructure. These results indicate that the auxiliary task improves the model’s ability to differentiate between visually similar surfaces with different semantic roles, resulting in more spatially consistent burned-area predictions. Overall, the findings demonstrate that auxiliary-task learning improves performance relative to a single-task multimodal model and serves as an effective mechanism for enhancing spatial segmentation quality.

4.12 Performance Across Fire Sizes

The impact of auxiliary-task learning varies across events and is strongly influenced by fire size. Figure 5 shows the mean ΔIoU between the multimodal multi-task model and its single-task counterpart, grouped by ground-truth burned-area fraction. The largest improvements occur for medium to large fires, particularly in the 0.05–0.20 burned-fraction range, where auxiliary supervision leads to consistent performance gains.

For very small fires, the contribution of the auxiliary task is limited. In these cases, stochastic ignition behavior and early suppression actions reduce the role of environmental factors, limiting the benefit of additional semantic context. As fires grow beyond initial containment, however, landscape structure and fuel continuity increasingly influence fire spread, allowing the multi-task model to better exploit land-cover information. The performance drop observed in the largest burned-area bin is attributed to the small number of samples and higher variability in extreme events, rather than a systematic limitation of the approach.

Overall, these results indicate that auxiliary-task learning is most effective for fires with sufficient spatial extent, where environmental context plays a dominant role in shaping burn patterns, highlighting its relevance for high-impact wildfire prediction.

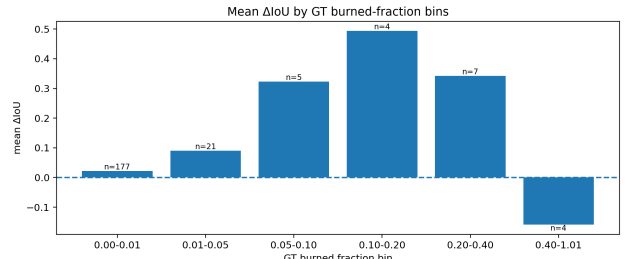


Figure 5: Mean ΔIoU between the multimodal multi-task and single-task models, stratified by ground-truth burned-area fraction. Auxiliary-task learning provides the largest gains for medium and large fires, where environmental structure dominates fire spread dynamics.

4.13 Modality Importance and Ablation Analysis

To address **RQ3**, we assess the relative contribution of each data modality using an inference-time ablation study, in which individual inputs are removed from the full multimodal model and the resulting performance drop is measured (Figure 6). Excluding Sentinel-2 imagery leads to the largest degradation in performance, with decreases of $\Delta\text{IoU} = -0.36$ and $\Delta\text{F1} = -0.53$, confirming that high-resolution optical data are the primary driver of spatial accuracy in burned-area segmentation.

Removing auxiliary modalities results in smaller but consistent decreases in both IoU and F1-score, indicating that each source provides complementary information. Among these inputs, meteorological variables from ERA5 cause a larger performance drop than static topographic features, with decreases of $\Delta\text{IoU} = -0.20$ and $\Delta\text{F1} = -0.33$, highlighting the importance of atmospheric conditions in shap-

ing fire spread patterns beyond what can be inferred from terrain alone. Topographic and anthropogenic features, while less influential individually, still help constrain predictions by encoding terrain structure and human-related boundaries.

Landsat-8 imagery provides a modest but reliable improvement, likely by supplementing Sentinel-2 observations during periods of reduced temporal coverage. Overall, the ablation results show that although optical imagery dominates performance, combining multiple environmental data sources leads to more robust early wildfire predictions, supporting the effectiveness of multimodal fusion.

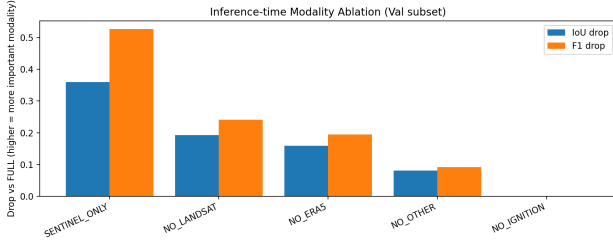


Figure 6: Inference-time modality ablation showing performance degradation relative to the full multimodal model. Larger drops indicate greater contribution to segmentation performance.

5 Conclusion

This study shows that the final burned-area extent of a wildfire can be predicted at the moment of ignition using multimodal pre-fire environmental data. By framing the problem as a semantic segmentation task and combining satellite imagery, meteorological variables, and topographic information within a single deep learning framework, the proposed method substantially outperforms single-modal and single-task baselines. At a conservative operating threshold, the multimodal model achieves an IoU of approximately 0.40, corresponding to a relative improvement of about 70%, and produces reliable spatial predictions at early stages of wildfire development.

The results also demonstrate that auxiliary land-cover supervision plays an important role in improving spatial consistency and reducing false positives, particularly in heterogeneous landscapes. Multi-task learning helps guide the model toward more realistic burned-area predictions and provides the greatest benefit for fires with sufficient spatial extent. Inference-time ablation experiments further confirm that high-resolution optical imagery is the primary source of predictive power, while meteorological and topographic inputs supply additional context that improves model robustness.

Overall, these findings indicate that combining multimodal data with auxiliary semantic supervision enables accurate and practical burned-area estimation at ignition time. Such early spatial forecasts can support more effective wildfire response by helping prioritize suppression efforts. Future work will focus on applying the framework to additional regions, incorporating uncertainty estimates, and testing generalization across different fire regimes to support real-world deployment.

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